

!pip install pandas !pip install matplotlib !pip install seaborn !pip install seaborn

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import pandas as pd

df = pd.read_csv(r"C:\Users\Sahil\Desktop\OneDrive\文档\SAHIL\customer
churn .csv")
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
0	7590-VHVEG	Female	0	Yes	No	1
1	5575-GNVDE	Male	0	No	No	34
2	3668-QPYBK	Male	0	No	No	2
3	7795-CF0CW	Male	0	No	No	45
4	9237-HQITU	Female	0	No	No	2

	MultipleLines	InternetService	OnlineSecurity	...
0	No phone service	DSL	No	...
1	No	DSL	Yes	...
2	No	DSL	Yes	...
3	No phone service	DSL	Yes	...
4	No	Fiber optic	No	...

	TechSupport	StreamingTV	StreamingMovies	Contract
0	No	No	No	Month-to-month
1	No	No	No	One year
2	No	No	No	Month-to-month
3	Yes	No	No	One year

No				
4	No	No	No	Month-to-month
Yes				

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

df.info(1)

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null	Count	Dtype
0	customerID	7043	non-null	object
1	gender	7043	non-null	object
2	SeniorCitizen	7043	non-null	int64
3	Partner	7043	non-null	object
4	Dependents	7043	non-null	object
5	tenure	7043	non-null	int64
6	PhoneService	7043	non-null	object
7	MultipleLines	7043	non-null	object
8	InternetService	7043	non-null	object
9	OnlineSecurity	7043	non-null	object
10	OnlineBackup	7043	non-null	object
11	DeviceProtection	7043	non-null	object
12	TechSupport	7043	non-null	object
13	StreamingTV	7043	non-null	object
14	StreamingMovies	7043	non-null	object
15	Contract	7043	non-null	object
16	PaperlessBilling	7043	non-null	object
17	PaymentMethod	7043	non-null	object
18	MonthlyCharges	7043	non-null	float64
19	TotalCharges	7043	non-null	object
20	Churn	7043	non-null	object

dtypes: float64(1), int64(2), object(18)

memory usage: 1.1+ MB

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")
df["TotalCharges"] = df["TotalCharges"].astype("float")
```

replacing blank with 0 as tenure is 0 and no total charges are recorded

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 7043 entries, 0 to 7042
```

```
Data columns (total 21 columns):
```

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2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	object
4	Dependents	7043 non-null	object
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	object
7	MultipleLines	7043 non-null	object
8	InternetService	7043 non-null	object
9	OnlineSecurity	7043 non-null	object
10	OnlineBackup	7043 non-null	object
11	DeviceProtection	7043 non-null	object
12	TechSupport	7043 non-null	object
13	StreamingTV	7043 non-null	object
14	StreamingMovies	7043 non-null	object
15	Contract	7043 non-null	object
16	PaperlessBilling	7043 non-null	object
17	PaymentMethod	7043 non-null	object
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	object
20	Churn	7043 non-null	object

```
dtypes: float64(1), int64(2), object(18)
```

```
memory usage: 1.1+ MB
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```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
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4	Dependents	7043 non-null	object
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	object
7	MultipleLines	7043 non-null	object
8	InternetService	7043 non-null	object
9	OnlineSecurity	7043 non-null	object
10	OnlineBackup	7043 non-null	object
11	DeviceProtection	7043 non-null	object

```

12 TechSupport      7043 non-null object
13 StreamingTV      7043 non-null object
14 StreamingMovies  7043 non-null object
15 Contract         7043 non-null object
16 PaperlessBilling 7043 non-null object
17 PaymentMethod    7043 non-null object
18 MonthlyCharges   7043 non-null float64
19 TotalCharges     7043 non-null object
20 Churn            7043 non-null object

```

```
dtypes: float64(1), int64(2), object(18)
```

```
memory usage: 1.1+ MB
```

```
df.isnull()
```

	customerID	gender	SeniorCitizen	Partner	Dependents		
tenure \							
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
7038	False	False	False	False	False	False	False
7039	False	False	False	False	False	False	False
7040	False	False	False	False	False	False	False
7041	False	False	False	False	False	False	False
7042	False	False	False	False	False	False	False

	PhoneService	MultipleLines	InternetService
OnlineSecurity ... \			
0	False	False	False
False ...			
1	False	False	False
False ...			
2	False	False	False
False ...			
3	False	False	False
False ...			
4	False	False	False
False ...			

...	...	...	...	...	...
7038	False	False	False	False	False
7039	False	False	False	False	False
7040	False	False	False	False	False
7041	False	False	False	False	False
7042	False	False	False	False	False
DeviceProtection	TechSupport	StreamingTV	StreamingMovies		
Contract \					
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
7038	False	False	False	False	False
7039	False	False	False	False	False
7040	False	False	False	False	False
7041	False	False	False	False	False
7042	False	False	False	False	False
PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges		
Churn					
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False

```
False
...
...
...
7038      False      False      False      False
False
7039      False      False      False      False
False
7040      False      False      False      False
False
7041      False      False      False      False
False
7042      False      False      False      False
False
```

```
[7043 rows x 21 columns]
```

```
df.isnull().sum()
```

```
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    0
Churn           0
```

```
dtype: int64
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470

min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
df.isnull().sum()
```

```
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    0
Churn           0
dtype: int64
```

```
df.duplicated()
```

```
0      False
1      False
2      False
3      False
4      False
...
7038   False
7039   False
7040   False
7041   False
7042   False
Length: 7043, dtype: bool
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

```
# converted the 0 and 1 values of senior citizen to yes/no to make it easier to understand
```

```
def conv(value):  
    if value == 1:  
        return "yes"  
    else:  
        return "no"
```

```
df['SeniorCitizen'] = df["SeniorCitizen"].apply(conv)
```

```
File <string>:6
```

```
df['SeniorCitizen'] = df["SeniorCitizen"].apply(conv)
```

```
IndentationError: unindent does not match any outer indentation level
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
PhoneService \						
0	7590-VHVEG	Female	0	Yes	No	1
No						
1	5575-GNVDE	Male	0	No	No	34
Yes						
2	3668-QPYBK	Male	0	No	No	2
Yes						
3	7795-CF0CW	Male	0	No	No	45
No						
4	9237-HQITU	Female	0	No	No	2
Yes						

	MultipleLines	InternetService	OnlineSecurity	...
DeviceProtection \				
0	No phone service	DSL	No	...
No				
1	No	DSL	Yes	...
Yes				
2	No	DSL	Yes	...
No				
3	No phone service	DSL	Yes	...
Yes				
4	No	Fiber optic	No	...
No				

	TechSupport	StreamingTV	StreamingMovies	Contract
PaperlessBilling \				
0	No	No	No	Month-to-month
Yes				
1	No	No	No	One year

No				
2	No	No	No	Month-to-month
Yes				
3	Yes	No	No	One year
No				
4	No	No	No	Month-to-month
Yes				

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

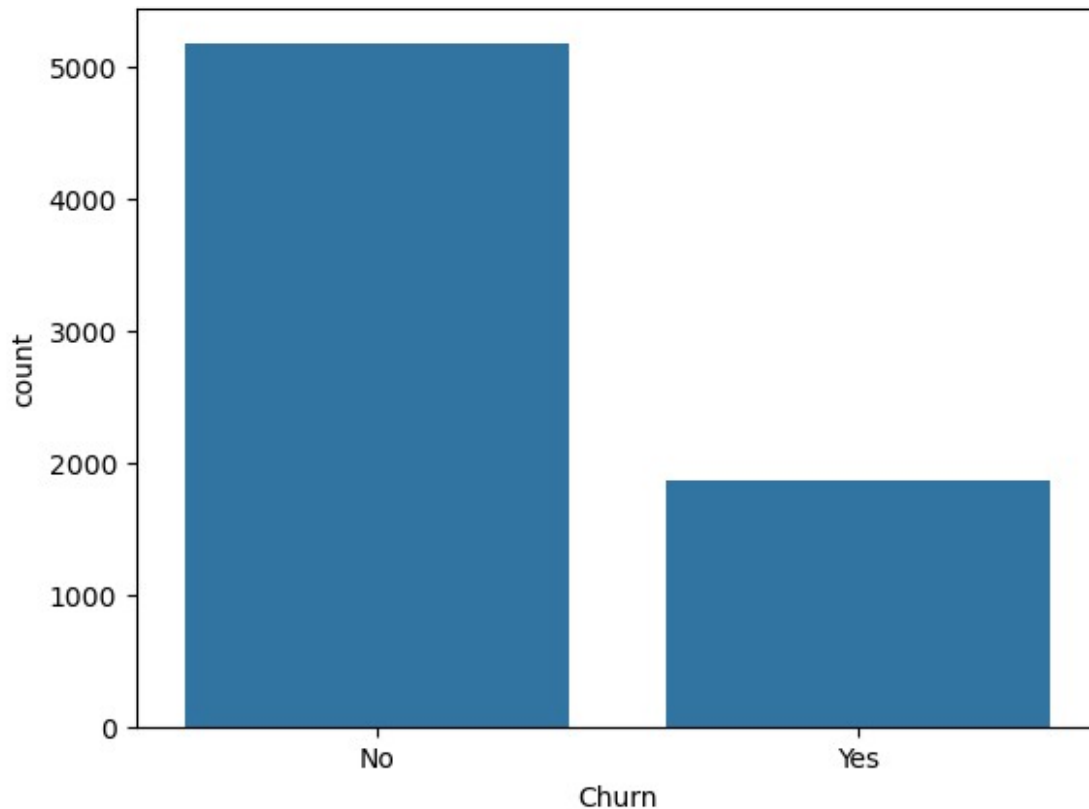
from the given pie chart we can conclude that 26.54% of our customers hvae churned out. # not let's explore the reason behind it

```
sns.countplot(x = 'Churn', data = df)
ax.bar_label (ax.containers[0])
plt.show()
```

```
-----
-----
NameError                                Traceback (most recent call
last)
```

```
Cell In[9], line 2
      1 sns.countplot(x = 'Churn', data = df)
----> 2 ax.bar_label (ax.containers[0])
      3 plt.show()
```

NameError: name 'ax' is not defined



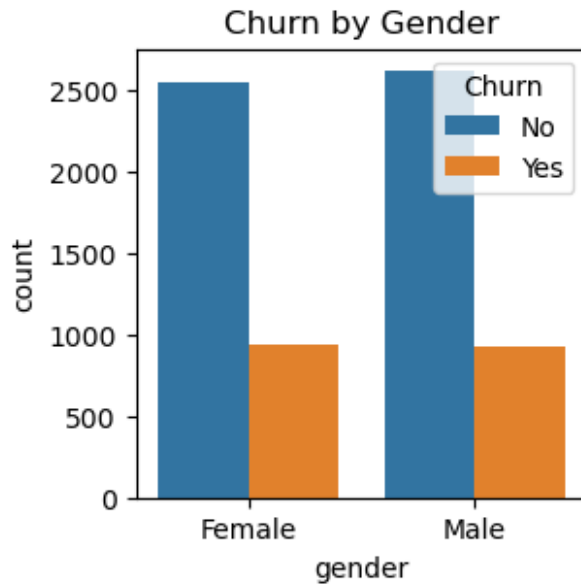
```
plt.figure(figsize = (3,4))
gb = df.groupby("Churn").agg({'Churn':"Count"})
plt.pie(gb['Churn'],labels = gb.index, autopct = "%1.2f%%")
plt.title("percentage of churned customers", fontsize = 10)
plt.show() nahi hua ye question
```

Cell In[10], line 5

```
plt.show() nahi hua ye question
^
```

SyntaxError: invalid syntax

```
plt.figure(figsize = (3,3))
sns.countplot(x = "gender", data = df, hue = "Churn")
plt.title("Churn by Gender")
plt.show()
```



senior citizen pychart ready upper wali ko dekh kar and ye bhi mujhse nahi hua hai

i want to create a stack bar chart which give me labels as % to total churn by senior citizen (stacked bar chart)

```
plt.figure(figsize=(5,4))
sns.countplot(x = "SeniorCitizen", hue="Churn")
plt.title("Churn by SeniorCitizen")
plt.show()
```

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Create percentage table
```

```
churn_pct = pd.crosstab(
    df["SeniorCitizen"],
    df["churn"],
    normalize="index"
) * 100
```

```
# Plot 100% stacked bar
```

```
ax = churn_pct.plot(
    kind="bar",
    stacked=True,
    figsize=(6, 4)
)
```

```
plt.title("Churn by SeniorCitizen (%)")
plt.xlabel("SeniorCitizen")
plt.ylabel("Percentage (%)")
plt.xticks(rotation=0)
```

```
plt.legend(title="Churn")

# Add percentage labels
for container in ax.containers:
    labels = [
        f"{bar.get_height():.1f}%" if bar.get_height() > 0 else ""
        for bar in container
    ]
    ax.bar_label(container, labels=labels, label_type="center")

plt.tight_layout()
plt.show()
```

```
-----
-----
NameError                                Traceback (most recent call
last)
```

```
Cell In[3], line 2
      1 plt.figure(figsize=(5,4))
----> 2 sns.countplot(x = "SeniorCitizen", hue="Churn")
      3 plt.title("Churn by SeniorCitizen")
      4 plt.show()
```

NameError: name 'sns' is not defined

<Figure size 500x400 with 0 Axes>

senior citizen pychart data ye abhi nahi hua hai
churn by senior citizen percentage

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Count Churn by SeniorCitizen
```

```
ct = pd.crosstab(["SeniorCitizen"], ["Churn"])
```

```
# Convert counts into percentages within each SeniorCitizen group
```

```
pct = ct.div(ct.sum(axis=1), axis=0) * 100
```

```
# Plot 100% stacked bar chart
```

```
ax = pct.plot(
    kind="bar",
    stacked=True,
    figsize=(7, 5),
    color=["#4CAF50", "#F44336"]
)
```

```
plt.title("Churn by SeniorCitizen (%)")
```

```
plt.xlabel("Senior Citizen")
```

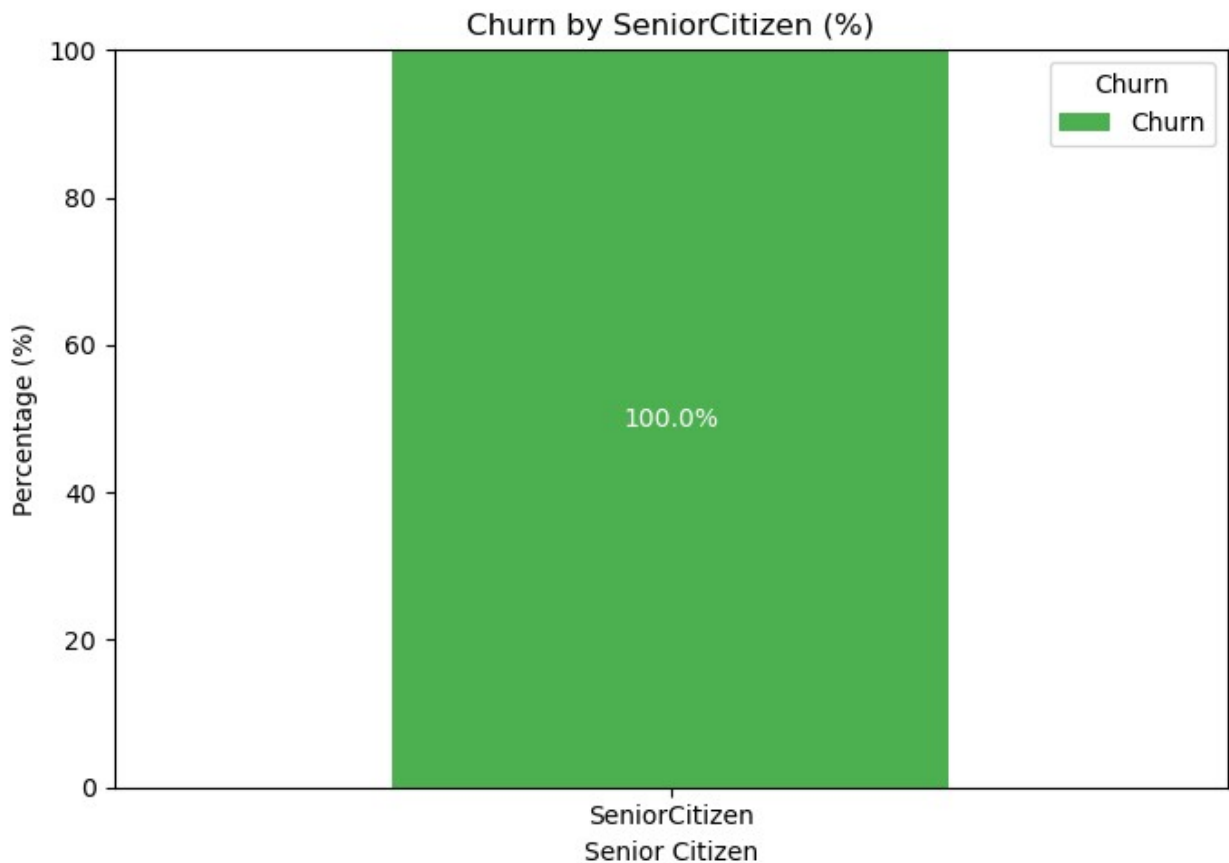
```

plt.ylabel("Percentage (%)")
plt.xticks(rotation=0)
plt.ylim(0, 100)
plt.legend(title="Churn")

# Add percentage labels
for container in ax.containers:
    labels = [
        f"{bar.get_height():.1f}%" if bar.get_height() > 0 else ""
        for bar in container
    ]
    ax.bar_label(container, labels=labels, label_type="center",
color="white")

plt.tight_layout()
plt.show()

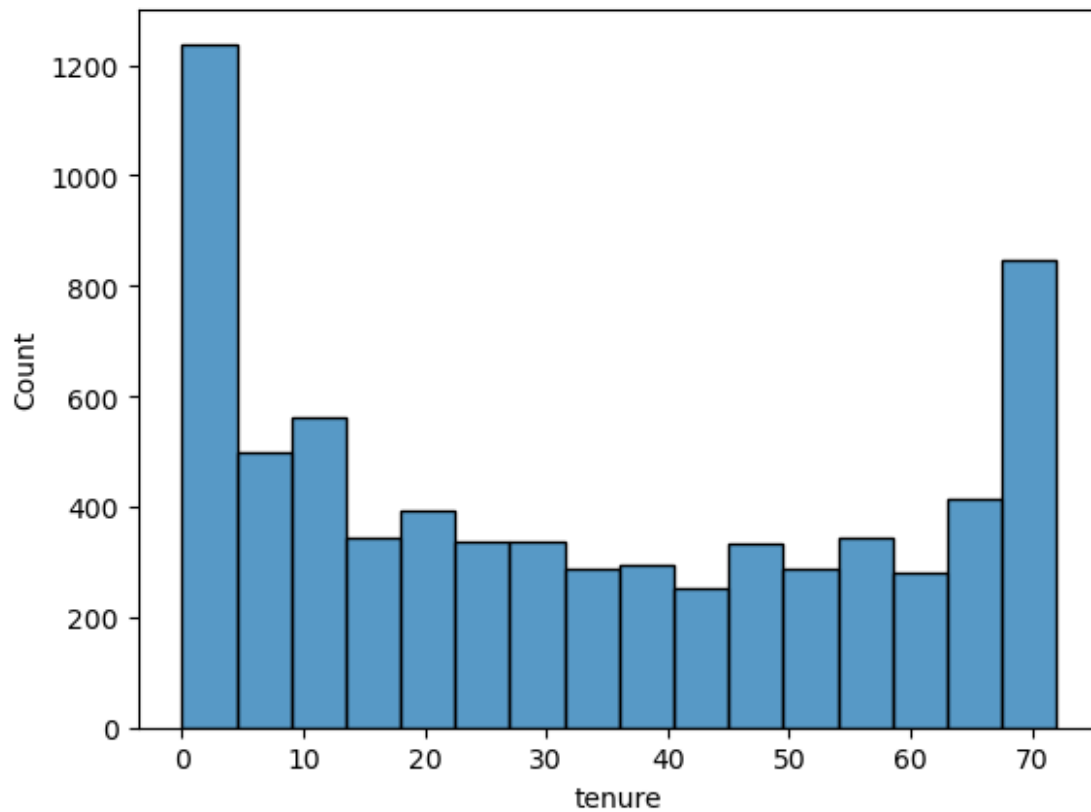
```



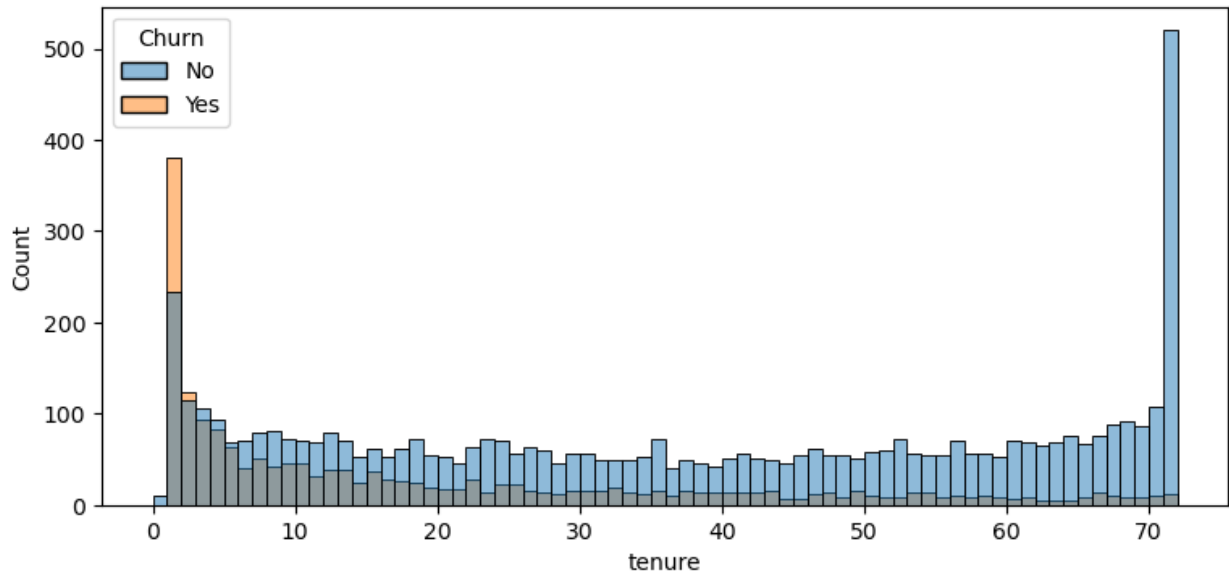
comperative is a greated percentage of people in senior citizen category have churned

count of senior citizen abhi ye rahe gya hai chart bana na reha gya hai

```
sns.histplot(x = "tenure", data = df)
plt.show()
```



```
plt.figure(figsize = (9,4))
sns.histplot(x = "tenure", data = df, bins = 72 , hue = "Churn")
plt.show()
```



people who have used our services for a long time have stayed and people who have used our services #1 or 2 Months have churned

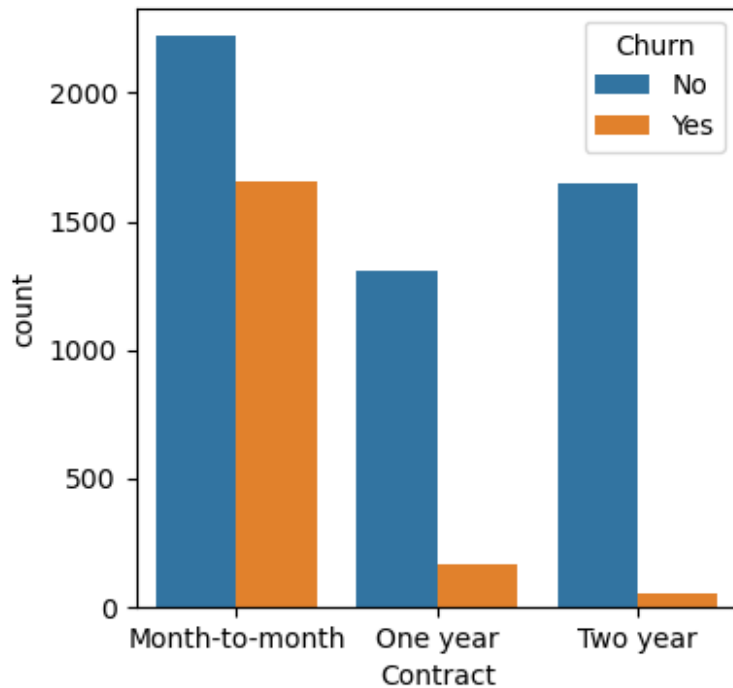
```
plt.figure(figsize = (4,4))
ax = sns.countplot(x = "Contract", data = df , hue = "Churn")
ax.bar_label(ax.Containers[0])
plt.title("Count of Customers by Contract")
plt.show()
```


 AttributeError Traceback (most recent call last)

Cell In[38], line 3

```
1 plt.figure(figsize = (4,4))
2 ax = sns.countplot(x = "Contract", data = df , hue = "Churn")
----> 3 ax.bar_label(ax.Containers[0])
      4 plt.title("Count of Customers by Contract")
      5 plt.show()
```

AttributeError: 'Axes' object has no attribute 'Containers'



people who have month to month contract are likely to churn then from those who have 1 or 2 year or contract

```
df.columns.values
```

```
array(['customerID', 'gender', 'SeniorCitizen', 'Partner',  
      'Dependents',  
      'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
      'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
      'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',  
      'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',  
      'TotalCharges', 'Churn'], dtype=object)
```

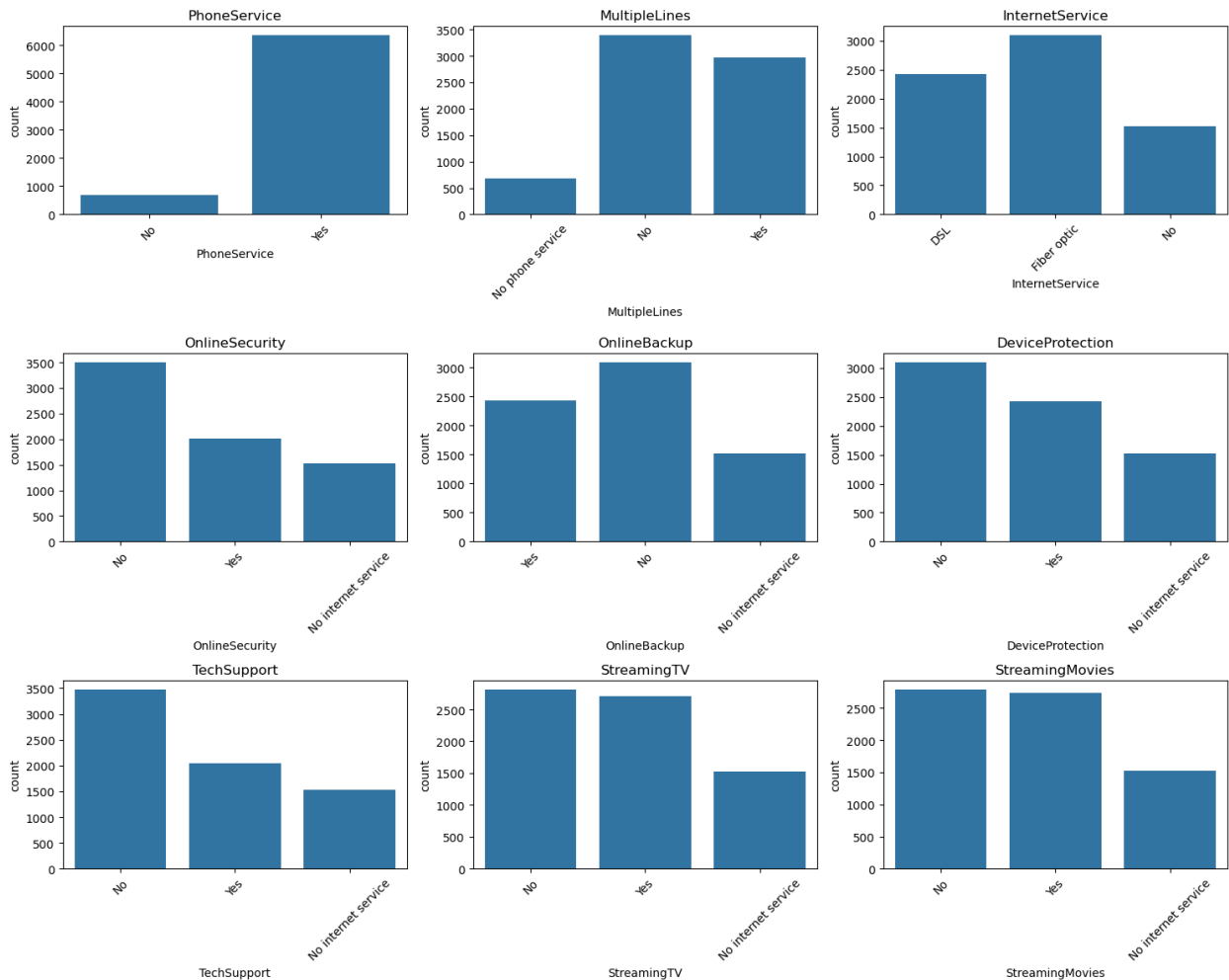
```
cols = [  
    'PhoneService',  
    'MultipleLines',  
    'InternetService',  
    'OnlineSecurity',  
    'OnlineBackup',  
    'DeviceProtection',  
    'TechSupport',  
    'StreamingTV',  
    'StreamingMovies'  
]
```

```
plt.figure(figsize=(15, 12))
```

```
for i, col in enumerate(cols, 1):
```

```
plt.subplot(3, 3, i)
sns.countplot(x=col, data=df)
plt.title(f'{col}')
plt.xticks(rotation=45)
```

```
plt.tight_layout()
plt.show()
```



```
# List of columns for which we want to create to count plots
Columns = ['PhoneService', 'MultipleLines ', ' InternetService ',
'OnlineSecurity',
'OnlineBackup', 'DeviceProtection', 'TechSupport',
'StreamingTV ', ' StreamingMovies ']

# Number of columns for the subplot grid (you can change this)
n_cols = 3
n_rows = (len(columns) + n_cols - 1) // n_cols # calculate number of
rows needed
```

```

# Create Subplots
fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, n_rows * 4)) #
adjust figsize as needed

# Flatten the axes array for easy iteration (handles both 1D and 2D
arrays)
axes = axes.flatten()

# Iterate over columns and plot count plots
for i, col in enumerate(columns):
    sns.countplot(x=col, data=df, ax=axes[i] , hue = df["chur"])
    axes[i].set_title(f'Count plot of {col}')
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Count')
# Remove empty subplots (if any)
for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

```

```

-----
-----
NameError                                Traceback (most recent call
last)
Cell In[50], line 7
      5 # Number of columns for the subplot grid (you can change this)

      6 n_clos = 3
----> 7 n_rows = (len(columns) + n_cols - 1) // n_cols # calculate
number of rows needed
      9 # Create Subplots
     10 fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, n_rows *
4)) # adjust figsize as needed

```

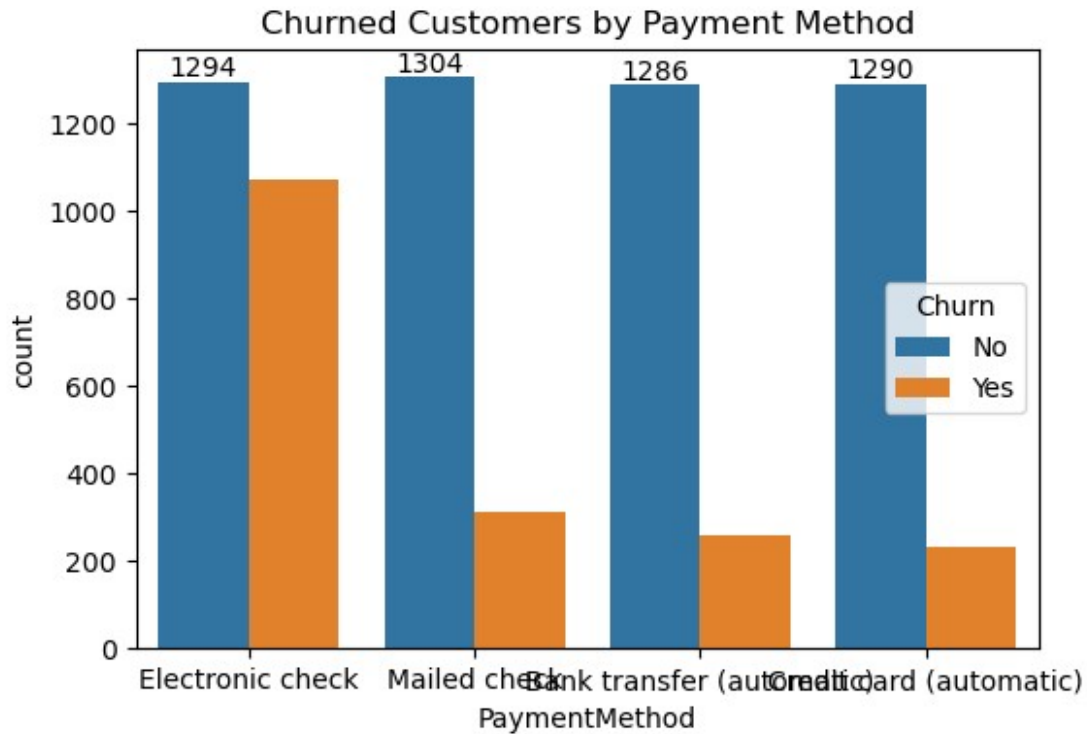
NameError: name 'columns' is not defined

The majority of customers who do **not** churn tend to have services like phone services, Internet services (Particularly DSL), Online security enabled, online backup, Techsupport, StreamingTv, Churnr rates are noticeably higher when these services are **not** used **or** unavailable

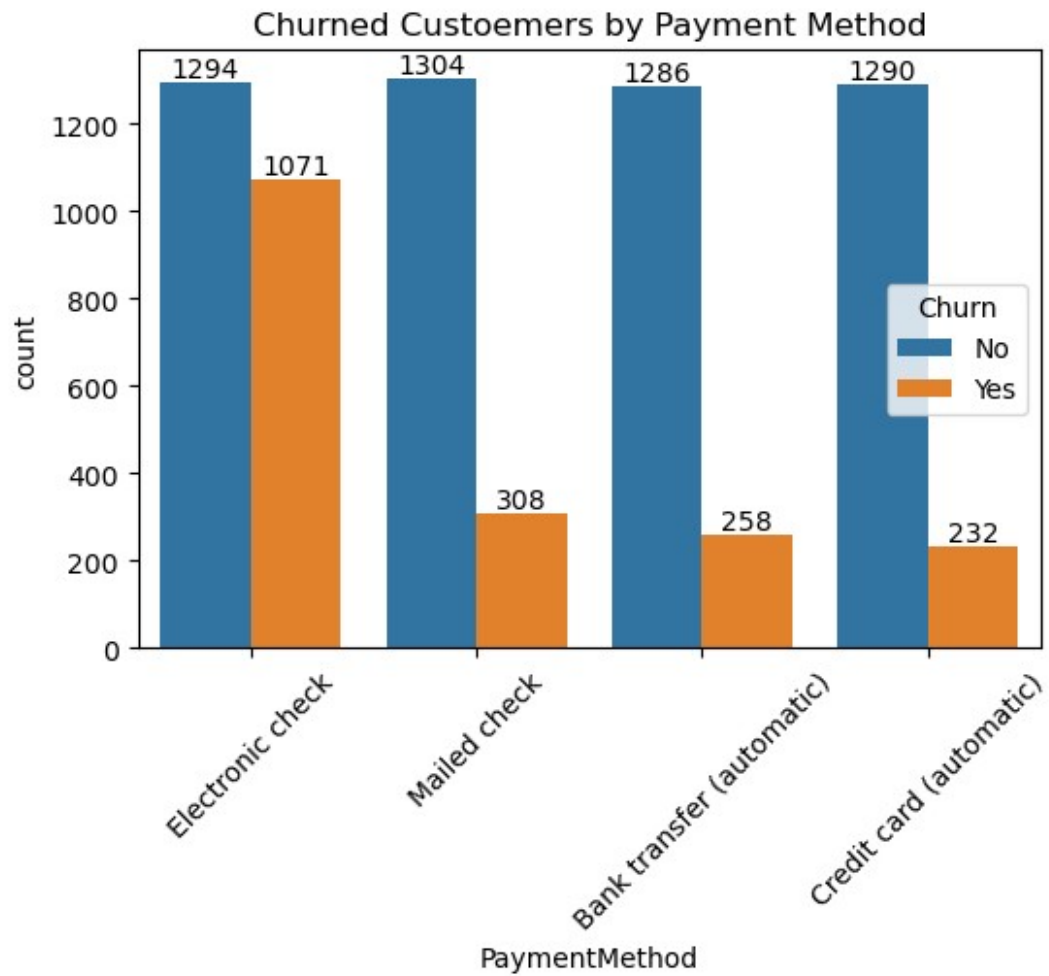
```

plt.figure(figsize = (6,4))
ax = sns.countplot(x = "PaymentMethod", data = df, hue = "Churn")
ax.bar_label(ax.containers[0])
plt.title ("Churned Customers by Payment Method")
plt.show()

```



```
plt.figure(figsize = (6,4))
ax = sns.countplot(x = "PaymentMethod", data = df, hue = "Churn")
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Churned Customers by Payment Method")
plt.xticks(rotation = 45)
plt.show()
```



Customer is likely to churn when he is using electronic check as a payment method.